Module - III

Fibre Optics

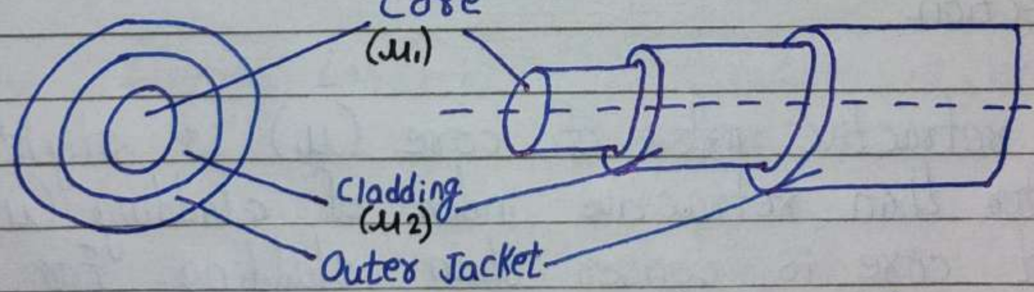
Introduction

Optical fibres are cables which can carry huge amount of information in the form of light from one place to another. The energy loss due to heating is negligible in optical fibres. They can sent data with higher transmission rate and wide bandwidth.

Fundamental Ideas About Optical Fibre

Basic Structure:

An optical fibre is a hair-thin, flexible, transparent cylindrical wire usually made up of pure glass or plastic. The light can propagate through it. The basic structure can be divided into three sections.



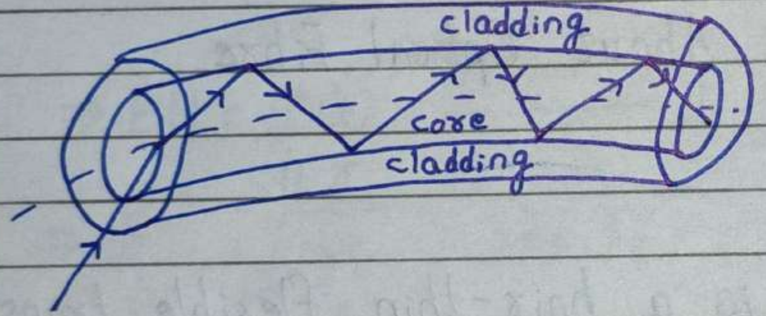
$[\mu_1 > \mu_2]$

(i) Core: It is the inner-most section made up of pure glass or plastic. It has diameter approximately 5-100 µm. The light travels through core.

(ii) Cladding: The core is surrounded by cladding made by glass or plastic. The refractive index and density of cladding is ~~slith~~ slightly lower than core. It has diameter about 125 μm.

(iii) Outer Jacket /Sheath: The outter-most section is made up of polymer. It protects the fibre from crussing, moisture, absorption, etc. The diameter of this is about 250 μm.

Imp. Basic Principle



Optical fibres are cylindrical wave-guide that propagates information, coded in the form of light wave along its axis. The basic principle is Total Internal Reflection.

The refractive index of core (μ_1) is slightly greater than refractive index of cladding (μ_2). So that core is denser than cladding. For a ray entering the fibre core, the angle of incidence is always greater than critical angle. The light suffers total internal reflection into the core.

Because of cylindrical symmetry in the structure of fibre, this ray will suffer TIR from lower surface also.

Therefore, the light get guided through the core by repeated TIR.

Fibre Classification

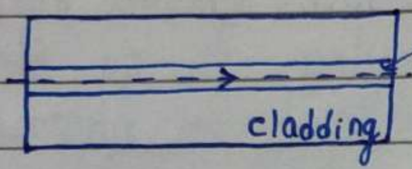
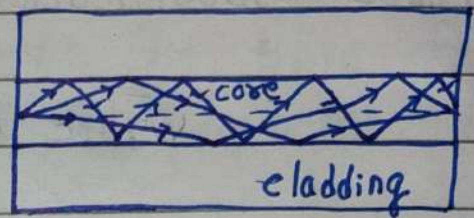
Fibre Classification

- Material
  - Glass
  - Plastic
- Modes
  - Single Node
  - Multi-node
- Refractive Index Profile
  - Step Index
  - Graded Index

On the basis of material

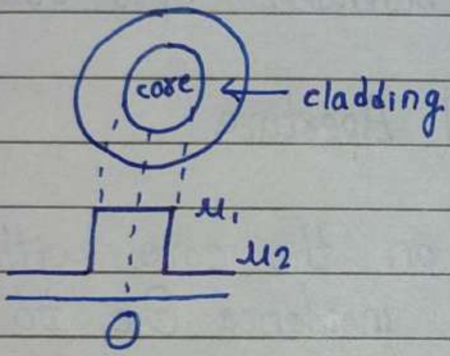
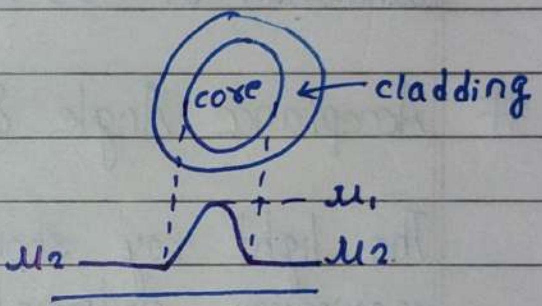
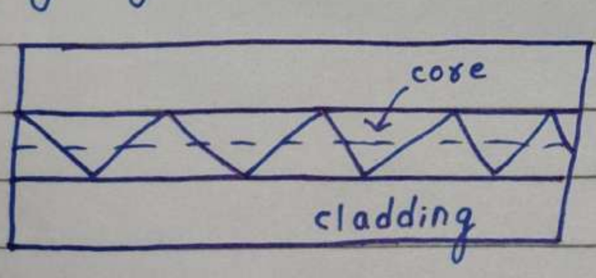
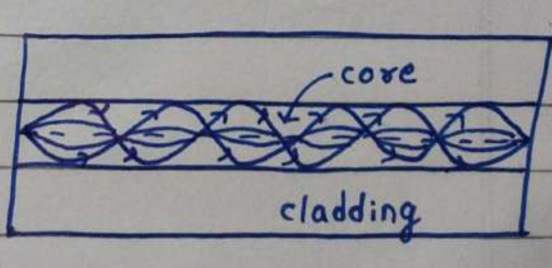
	Glass Fibre	Plastic Fibre
i.	Core and cladding are made up of glass.	i. Core and cladding are made up of plastic.
ii.	Attenution (Energy Loss) is very low.	ii. Attenution is comparatively higher.
iii.	Bandwidth is high.	iii. Bandwidth is low.
iv.	They are brittle and expensive.	iv. They are flexible and cheaper
v.	Used in long distance communication and for high speed data transmission.	v. Used in short distance, low speed networks and decorative lighting.

On the basis of Modes

	Single Mode (S.M.)		Multi-mode (M.M.)
i.	The diameter of core is very low, about 5-10 µm.	i.	The core diameter is large, about 10-100 µm.
ii.	 There is only a single path available (mode) for light to travel.	ii.	 There are many paths available for light to travel.
iii.	The difference between refractive index of core and cladding is very low.	iii.	The difference is large.
iv.	Acceptance angle and numerical aperture is small.	iv.	Acceptance angle and numerical aperture is large.
v.	Only lasers are used as light source.	v.	LED can be used as light source.
vi.	Single mode fibres are more expensive but efficient.	vi.	Multi-mode fibres are less expensive and less efficient.
vii.	Modal dispersion is very low.	vii.	Modal dispersion is large.

viii. They have high information carrying capability.	viii. They have low information carrying capability.
ix. Single Mode fibres are used in ~~short~~ short distance communication.	ix. Multi-mode fibres are used in long distance communication.

On the basis of Refractive Index

Step Index (S.I.)	Graded Index (GRIN)
i.  In S.I. fibre, the refractive index of core is constant.	i.  In GRIN fibre, the refractive index of core is highest (μ_1) at the centre and decreases in a parabolic manner, as we move towards the cladding.
ii. The light propagates in a zig-zag manner. 	ii. The light ray bends smoothly like a sine-wave. 

iii. Dispersion loss is high.	iii. Dispersion loss is low.
iv. Attenuation is higher.	iv. Attenuation is lower.
v. Acceptance angle and numerical apperture is large.	v. Acceptance angle and numerical apperture is small.
vi. $V < 2.405$	vi. $V > 2.405$
vii. S.I. fibre may have bandwidth of 50 MHz.	vii. GRIN fibre can have bandwidth upto 600 MHz.

Imp. # Acceptance Angle & Numerical Aperture

The light ray should incident on the core within a maximum external angle of incidence θ_e to fulfill the condition of T.I.R. for propagating down the core. This angle is called acceptance angle of the fibre. The value of acceptance angle depends upon core diametre and core material.

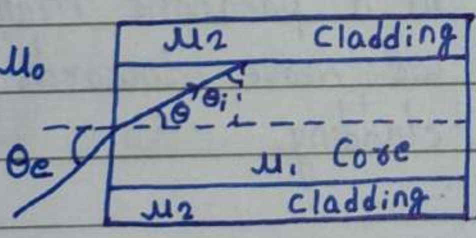


Fig.-1

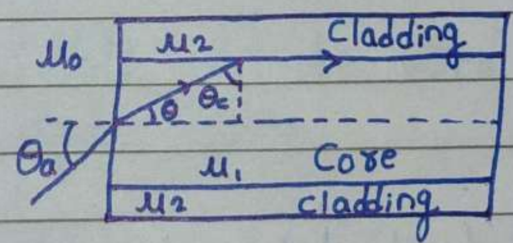


Fig.-2

According to Snell's Law,

$$\frac{\mu_1}{\mu_0} = \frac{\sin\theta_e}{\sin\theta}$$

$$\Rightarrow \mu_0 \sin\theta_e = \mu_1 \sin\theta$$

$$\because \theta + \theta_i = 90^\circ$$
$$\Rightarrow \theta = 90 - \theta_i$$

$$\mu_0 \sin\theta_e = \mu_1 \sin(90 - \theta_i)$$

$$\Rightarrow \mu_0 \sin\theta_e = \mu_1 \cos\theta_i$$

$$\Rightarrow \mu_0 \sin\theta_e = \mu_1 \sqrt{1 - \sin^2\theta_i} \quad \text{———} \quad (1)$$

If we increase θ_e, the angle of incidence at cladding θ_i will decrease.

In figure. (2), θ_e is increased so that θ_i becomes equal to critical angle θ_c. This value of θ_e is called Acceptance angle θ_a.

Putting $\theta_e = \theta_a$ & $\theta_i = \theta_c$ in eq^n. (1),

$$\mu_0 \sin\theta_a = \mu_1 \sqrt{1 - \sin^2\theta_c}$$

$$\because \sin\theta_c = \frac{\mu_2}{\mu_1}$$

$$\mu_0 \sin\theta_a = \mu_1 \sqrt{1 - \frac{\mu_2^2}{\mu_1^2}}$$

$$\Rightarrow \mu_0 \sin\theta_a = \sqrt{\mu_1^2 - \mu_2^2}$$

$$\Rightarrow \sin\theta_a = \frac{\sqrt{\mu_1^2 - \mu_2^2}}{\mu_0}$$

$$\Rightarrow \sin\theta_a = \frac{\sqrt{\mu_1^2 - \mu_2^2}}{\mu_0}$$

$$\Rightarrow \boxed{\theta_a = \sin^{-1}\left(\frac{\sqrt{\mu_1^2 - \mu_2^2}}{\mu_0}\right)}$$ [For Any Medium]

$$\boxed{\theta_a = \sin^{-1}\sqrt{\mu_1^2 - \mu_2^2}}$$ [For Air]

Numerical Aperture:

It represents the light gathering capacity of an optical fibre. When a light is emmited from a source situated at fibre axis near the end surface, the fraction of light that passes through core is equal to $(NA)^2$. The value of NA is given by:-

$$(NA)^2 = \frac{I}{I_0}$$

$$\boxed{NA = \sin\theta_a = \sqrt{\mu_1^2 - \mu_2^2}}$$

Relative R.I. Difference (Δ)

$$\Delta = \frac{\mu_1 - \mu_2}{\mu_1}$$

$$\Rightarrow \Delta = \frac{\mu_1 - \mu_2}{\mu_1} \times \frac{(\mu_1 + \mu_2)}{(\mu_1 + \mu_2)}$$

$$\Rightarrow \Delta = \frac{\mu_1^2 - \mu_2^2}{\mu_1(\mu_1 + \mu_2)}$$

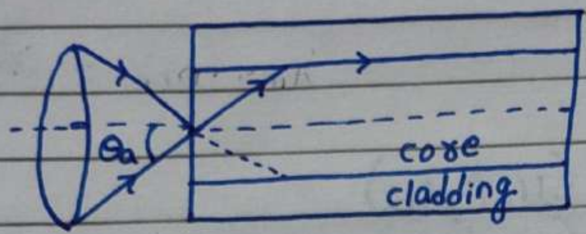
$$\because \mu_1 + \mu_2 \approx 2\mu_1$$

$$\Delta = \frac{\mu_1^2 - \mu_2^2}{2\mu_1^2}$$

$$\Rightarrow \mu_1^2 - \mu_2^2 = 2\mu_1^2 \Delta$$

$$\Rightarrow \boxed{NA = \mu_1 \sqrt{2\Delta}}$$

Acceptance Cone :



All the rays falling within the cone formed with acceptance angle θ_a as vertex angle would be transmitted through the fibre core via T.I.R.. This cone is called Acceptance Cone.

Normalized Frequency (V-number)

The various structural parameters of a fibre can be combined into a normalized parameter known as Normalized Frequency (V-number). It is also called Cutoff Parameter.

$$\boxed{V = \frac{\pi d}{\lambda}(NA)}$$

$$\Rightarrow \boxed{V = \frac{\pi d}{\lambda}\sin\theta_a}$$

$$\Rightarrow \boxed{V = \frac{\pi d}{\lambda}\sqrt{\mu_1^2 - \mu_2^2}}$$

$$\Rightarrow \boxed{V = \frac{\pi d}{\lambda}\mu_1\sqrt{2\Delta}}$$

[$d \rightarrow$ Diameter of Core
$\lambda \rightarrow$ Wavelength of Light]

No. of mode = $\frac{V^2}{2}$ [Step Index]

No. of mode = $\frac{V^2}{4}$ [Multi-mode Graded Index]

In Single Mode,
No. of mode = 1

For SM, $V < 2.405$
For MM, $V > 2.405$

Advantages of Optical Fibre

- Optical fibres have negligable energy loss as compare to metal wire.
- Light as a carrier-wave have a huge information carrying capacity
- They are light-weighted and small in size.
- Fibres are cheaper.
- Bandwidth of optical fibre is high.
- They have a longer lifetime.

Applications ~~Disadvantages~~ of Optical Fibre

- They are widely used in broadcasting of T.V. and cable T.V. network.
- They are used in remote monitoring & surveillance.
- They are frequently used for transmission of digital data such as computers.
- They are used as light-guide in medical field.
- They are used to formed sensors, to measure the physical and chemical parameters.

Numericals

Imp. Q) A silica-glass optical fibre has a core refractive index of 1.5 and cladding refractive index of 1.45. Calculate the numerical aperture, acceptance angle and critical angle?

Soln:

$\mu_1 = 1.5$

$\mu_2 = 1.45$

$\therefore NA = \sqrt{\mu_1^2 - \mu_2^2} = 0.38405$

$\theta_a = \sin^{-1}\sqrt{\mu_1^2 - \mu_2^2} = 22.585°$

$\theta_c = \sin^{-1}\left(\frac{\mu_2}{\mu_1}\right) = 75.1648°$

Q) If the fractional difference between core and cladding refractive indices of a fibre is 0.0135 and numerical aperture is 0.2425. Calculate the refractive index of core and cladding material?

Soln:

$\Delta = 0.0135$

$NA = 0.2425$

We have,

$NA = \mu_1\sqrt{2\Delta}$

$$\Rightarrow \mu_1 = \frac{NA}{\sqrt{2\Delta}} = 1.4758.$$

Also, we know,

$$(NA)^2 = \mu_1^2 - \mu_2^2$$

$$\Rightarrow \mu_2^2 = \mu_1^2 - (NA)^2$$

$$\Rightarrow \mu_2 = \sqrt{\mu_1^2 - (NA)^2} = 1.4557.$$

Q) A step index fibre has core refractive index 1.466 and cladding refractive index 1.460. If the operating wavelength of the ray is 0.85 um, calculate the cut-off parameter and the number of guided modes which the fibre will support? The diameter of core is 50 um.

Sol^n:

$\mu_1 = 1.466$
$\mu_2 = 1.460$
$\lambda = 0.85$ um
$d = 50$ um

We know,

$$V = \frac{\pi d}{\lambda}\sqrt{\mu_1^2 - \mu_2^2} = 24.485$$

$$\text{No. of modes} = \frac{V^2}{2} = 299.$$

Q) The core diameter of a multi-mode fibre is 70 μm and relative R.I. difference is 1.5%. If it operates at a wavelength 0.85 μm and refractive index of core is 1.46, calculate the refractive index of cladding, normalized frequency (V-number) and no. of guided modes?

Sol^n:

$d = 70\ \mu m$

$\Delta = 1.5\% = 0.015$

$\lambda = 0.85\ \mu m$

$\mu_1 = 1.46$

We have,

$$\Delta = \frac{\mu_1 - \mu_2}{\mu_1}$$

$$\Rightarrow 0.015 = \frac{1.46 - \mu_2}{1.46}$$

$$\Rightarrow 1.46 - \mu_2 = 0.0219$$

$$\Rightarrow \mu_2 = 1.46 - 0.0219 = 1.4381$$

Also,

$$V = \frac{\pi d}{\lambda}\ \mu_1 \sqrt{2\Delta} = 65.4248$$

No. of guided modes $= \frac{V^2}{2} = 2,140.$

Q) A step-index fibre has core refractive index 1.466 and cladding refractive index 1.460. Compute the maximum radius allowed for a fibre, if it is supported only one mode at a wavelength 1300 nm?

Sol^n:

$\mu_1 = 1.466$

$\mu_2 = 1.460$

$\lambda = 1300$ nm

For single mode,

$$V < 2.405$$

$$\Rightarrow \frac{\pi d}{\lambda}\sqrt{\mu_1^2 - \mu_2^2} < 2.405$$

$$\Rightarrow d < \frac{2.405 \times \lambda}{\pi\sqrt{\mu_1^2 - \mu_2^2}}$$

$$\Rightarrow d < 7510.965 \text{ nm}$$

$$\Rightarrow r < 3755.482 \text{ nm}$$

$$\Rightarrow r < 3.755 \ \mu m$$

$\therefore$ The maximum radius allowed is $3.755 \ \mu m$.